

Revised Pakistan-Local Budget & Prediction Framework

Priority 1 Forecasting Stack for a Rooftop PM2.5 Project

Core automated build	PKR 9,046
Core + validation tool	PKR 11,136
Main new additions	BME280 + rain-event sensing + wind validation strategy
External forecasting inputs	Public wind direction / wind speed / rainfall feed + time & activity features

This revised report shifts the project from simple PM monitoring to a forecasting-first architecture. The goal is not only to collect PM2.5 observations, but also to assemble a practical feature stack that can support predictive modelling and then convert the resulting forecasts into operational decisions for businesses and facilities.

1. Why the Build Was Revised

The earlier low-cost design measured PM2.5 and basic temperature/humidity. For forecasting, that is not enough. The revised Priority 1 design therefore adds barometric pressure, rain-event sensing, richer temporal variables, and a hybrid weather-data strategy. The on-site hardware captures the variables that are inexpensive to measure locally, while external hourly weather data supplies continuous wind information when low-cost local wind-direction hardware is not realistically available in the surfaced Pakistan-local search results.

2. Revised Priority 1 Hardware Budget

Item	Selected Option	Vendor	Price (PKR)
PM sensor (local surfaced laser PM sensor)	PM2.5/PM10 sensor	Daraz	4,589
Controller	ESP32 WROOM-32D development board/Hallroad Lahore / Epro		1,150
Environmental sensor	BME280 temp/humid/press sensor	Hallroad Lahore	650
Data logging module	Arduino microSD card reader module	Hallroad Lahore	147
Storage media	Samsung EVO Plus 16GB microSD	Daraz	989
Weather enclosure	Waterproof DIY enclosure	Hallroad Lahore	343
Jumper / hookup wires	20cm mixed jumper wire kit	Daraz	199
Power supply	5V 2A power supply with cable	Hallroad Lahore	294
Mounting accessory	100 pcs cable ties	Daraz	100
Mounting accessory	Pack of 2 stainless 2-inch hose clamps	Daraz	139
Radiation shield material	White plastic plate set	Daraz	299
Rainfall proxy sensor	Rain drops detection sensor module	Hallroad Lahore	147
TOTAL			9,046

Core automated build total: **PKR 9,046**. Optional handheld anemometer for field validation: **PKR 2,090**. Core plus validation total: **PKR 11,136**.

3. Hybrid Weather Data Strategy

The revised build uses a BME280 module to measure temperature, humidity, and pressure locally. A low-cost rain module is included to flag wet conditions or rain events. Continuous wind speed and wind direction are then added from the nearest reliable public weather feed. This hybrid approach is more practical than forcing the whole design to depend on hard-to-source low-cost local wind-direction hardware, and it still supports forecasting.

4. Full Prediction Framework

Target variable: PM2.5 concentration. Forecast horizons: 1-hour ahead, 3-hour ahead, 6-hour ahead, and 24-hour ahead. The feature set should include particulate lags (PM2.5/PM10/PM1), on-site meteorology (temperature, humidity, pressure), rain-event flag, public weather features (wind speed, wind direction, rainfall cross-check), and operational-context features such as time variables and campus activity logs.

Group	Feature	Source	Role
Target	PM2.5 concentration	PMS7003 PM2.5	Primary variable to forecast
Lag feature	PM2.5(t-1..t-24)	Derived from sensor log	Captures persistence / autocorrelation
Lag feature	PM10(t-1..t-24)	PMS7003 PM10	Useful companion particulate predictor
Lag feature	PM1(t-1..t-24)	PMS7003 PM1	Available from PMS7003 if logged
Meteorology	Temperature	BME280	Forecasting covariate
Meteorology	Relative humidity	BME280	Forecasting covariate
Meteorology	Pressure	BME280	Represents weather-system / dispersion changes
Meteorology	Rain event flag	Rain drop sensor	Cheap wetness proxy / cleansing event indicator
Meteorology	Wind speed	Public weather feed (handheld validation)	Dispersion / transport covariate
Meteorology	Wind direction	Public weather feed	Source-region / transport covariate
Temporal	Hour of day	Software-derived	Traffic and human activity pattern
Temporal	Day of week	Software-derived	Weekday / weekend pattern

5. Recommended Modelling Approach

Start with persistence and linear baselines, then progress to tree-based models such as Random Forest, XGBoost, or LightGBM. Use walk-forward validation instead of a random split, because air-quality data are time-ordered and operational forecasts must reflect future performance. If the dataset becomes large and clean enough, sequence models can be explored later, but they are not necessary for a strong student-level forecasting system.

- Step 1: Collect and synchronize data — Align PM, BME280, rain event, public weather, and manual logs at a common timestamp (e.g., hourly).
- Step 2: Quality control — Remove impossible values, fill short gaps carefully, mark long gaps, and check SD card integrity.
- Step 3: Feature engineering — Create lag features, rolling means, peak-hour flags, weekend flags, rain-event flags, and wind-direction bins.
- Step 4: Baselines — Start with persistence baseline and linear / ridge regression.
- Step 5: Tree models — Train Random Forest / XGBoost / LightGBM using engineered features.
- Step 6: Sequence models (optional) — Use LSTM/GRU only if you have enough clean hourly history and team capability.

- Step 7: Time-series validation — Use rolling-origin / walk-forward validation instead of random train-test split.
- Step 8: Metrics — Evaluate RMSE, MAE, R^2 , MAPE (carefully), and exceedance classification metrics if thresholds matter.
- Step 9: Interpretability — Use SHAP / feature importance to show which drivers matter most.
- Step 10: Operationalization — Convert forecast ranges into actions, alerts, and business playbooks.

6. Turning Forecasts into Operational Decisions

The project goal includes operational use. Therefore, forecast outputs should be mapped to business actions. The table below is a starting operational playbook.

Business Context	Trigger	Action
Facilities / campuses	Forecast PM2.5 high in next 3–6 h	Increase indoor filtration, reduce fresh-air intake if appropriate, reschedule
Construction / site work	High PM2.5 + low wind dispersion	Reduce dust-generating activity windows, increase wet suppression, mo
Logistics / delivery hubs	Morning peak forecast	Shift loading/unloading and route dispatch to lower-exposure periods wh
Retail / hospitality	Poor air quality episode	Adjust HVAC settings, display indoor-air messaging, protect staff in outd
ESG / HSE reporting	Repeated exceedance risk	Use alerts for incident logs, mitigation records, and business continuity p

7. Deployment Notes

Because the revised design is forecasting-first, the most important practical requirements are timestamp consistency, data quality control, and a clear decision layer. The team should maintain one synchronized hourly dataset, include feature engineering for lags and rolling windows, and preserve logs of special events such as generator operation, road congestion, or construction work.

8. Source Log

A structured source log with local pricing references is included in the updated Excel workbook. The principal reference IDs used for the revised build are: turn8search57, turn8search23, turn11search156, turn8search32, turn8search11, turn8search17, turn9search85, turn8search42, turn9search81, turn9search95, turn9search86, turn11search151, and turn11search164.